

Sectoral Diversification and Knowledge-Economy Efficiency

A replication of Siddiqui and Afzal for the UAE with an Ecuador extension

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1 Abstract

This article replicates and extends the empirical strategy used by Siddiqui and Afzal to evaluate sectoral diversification and knowledge-economy performance in the United Arab Emirates. The replication has two parts. First, it recomputes the Herfindahl-Hirschman Index (HHI) for gross fixed capital formation by sector and compares the result with the values published in the original article. Second, it audits the article's DEA CCR exercise and shows that the published pattern of universal efficiency is consistent with a low-discrimination DEA design.

The extension applies the same diversification logic to Ecuador using value added by seven ISIC branches from the UNSD National Accounts Main Aggregates. It also implements an exploratory DEA model for Ecuador in which years are DMUs. The evidence supports three conclusions: the UAE HHI replication is robust; the original DEA design is underpowered; and Ecuador is better described as moderately diversified when the sectoral partition has enough detail.

2 Introduction

Diversification is central to the policy discussion in hydrocarbon-dependent and commodity-exposed economies. The article replicated here studies the UAE’s transition toward a knowledge-based economy using descriptive indicators, HHI concentration measures, and DEA efficiency scores. This replication keeps that structure but separates descriptive replication from methodological audit.

The contribution is practical. A small difference in HHI values does not change the UAE classification. By contrast, a small number of DEA decision-making units relative to the number of inputs and outputs can change the interpretation of efficiency results. The extension to Ecuador adds a second warning: if sectors are aggregated too coarsely, the HHI can mechanically imply concentration before the data are analyzed.

3 Data and Reproducibility

The project uses four data blocks. The source inventory below is generated from the replication pipeline.

	dataset	file	source	use
0	UAE GFCF by sector	data/raw/hhi/gfcf_sector	KAPSARC Data Portal / FCSC, originally Bayanat...	UAE HHI replication
1	Published UAE HHI values	data/processed/hhi_values	Siddiqui, paper Aftab (2022), Table 3	Replication benchmark
2	World Development Indicators	data/raw/wdi/wdi_uae_2000-2020.csv	World Bank WDI API	UAE DEA audit
3	UNSD National Accounts Main Aggregates	outputs/tables/hhi_ecuador.csv	UNSD AMA API, GDP and breakdown at current prices	Ecuador HHI, year-level DEA, and sector-level DEA

The UAE HHI replication uses gross fixed capital formation by 18 sectors for 2010-2017. Published HHI values from the source article are stored separately so the comparison is transparent. The UAE DEA audit uses World Development Indicators for 2000-2020. The Ecuador extension uses the UNSD National Accounts Main Aggregates and constructs a mutually exclusive seven-branch value-added partition.

4 Methods

4.1 HHI

For each year, the HHI is computed as:

$$HHI_t = \sum_i \left(\frac{X_{it}}{X_t} \right)^2$$

where (X_{it}) is the value of sector or branch (i) in year (t), and (X_t) is the total across sectors or branches in that year. The classification thresholds follow the replicated article:

- $HHI < 0.15$: diversified
- $0.15 \leq HHI < 0.25$: moderately diversified
- $HHI \geq 0.25$: least diversified

4.2 DEA CCR

The project implements an input-oriented CCR DEA model. For each DMU (o), the linear program minimizes θ , subject to:

$$X\lambda \leq \theta x_o, \quad Y\lambda \geq y_o, \quad \lambda \geq 0$$

An efficiency score of 1 indicates that the DMU lies on the estimated frontier. DEA results are fragile when there are too few DMUs relative to model dimensions. The audit therefore applies the adequacy rule:

$$n_{DMU} \geq 3(m + s)$$

where (m) is the number of inputs and (s) is the number of outputs. Models below the threshold are reported as under-specified.

5 UAE HHI Replication

	year	replicated_hhi	published_hhi	abs_diff	classification
0	2010	0.2364	0.2367	0.0003	Moderately diversified
1	2011	0.2366	0.2364	0.0002	Moderately diversified
2	2012	0.2479	0.2477	0.0002	Moderately diversified
3	2013	0.2432	0.2429	0.0003	Moderately diversified
4	2014	0.2431	0.2428	0.0003	Moderately diversified
5	2015	0.2492	0.2489	0.0003	Moderately diversified
6	2016	0.2446	0.2446	0.0000	Moderately diversified
7	2017	0.2468	0.2445	0.0023	Moderately diversified

The recomputed HHI closely matches the published Table 3 values. In seven of eight years the absolute difference is no greater than 0.0003. The largest difference occurs in 2017, where the recomputed value is 0.2468 and the published value is 0.2445. The difference does not change the substantive classification: every year remains in the moderately diversified interval.

	metric	value
0	Maximum absolute difference	0.00230
1	Mean absolute difference	0.00049

6 UAE DEA Audit

	model	n_inputs	n_outputs	n_dm	threshold_3x	verdict	n_efficient	mean_efficiency
0	ICT_Knowledge_Diffusion	2	3	11	9	VALIDO	1	0.4183
1	Innovation_to_GDP	1	1	11	9	VALIDO	2	0.9410
2	Diversification_Efficiency	3	3	21	12	VALIDO	1	0.5181
3	Table2_Macro_Degenerate	3	4	6	21	SUBE-SPECIFICADO	6	1.0000

The model closest to the original DEA specification uses three inputs and four outputs but has only six complete annual DMUs. Its adequacy threshold is 21 DMUs, so it is heavily under-specified. That model classifies all six years as efficient, reproducing the paper’s universal-efficiency pattern but not providing strong evidence that all knowledge-economy pillars are efficient.

The redesigned UAE models satisfy the adequacy rule. They no longer produce universal efficiency: the number of efficient years ranges from one to two, and mean efficiency ranges from 0.4183 to 0.9410. This supports the interpretation that the original DEA finding is mainly a discrimination-power problem.

7 Ecuador Extension

7.1 HHI by ISIC Branch

	year	hhi	classification
0	2000	0.1824	Moderately diversified
1	2001	0.1853	Moderately diversified
2	2002	0.1932	Moderately diversified
3	2003	0.2008	Moderately diversified
4	2004	0.2010	Moderately diversified
5	2005	0.1962	Moderately diversified
6	2006	0.1949	Moderately diversified
7	2007	0.1978	Moderately diversified
8	2008	0.1962	Moderately diversified
9	2009	0.2023	Moderately diversified
10	2010	0.2000	Moderately diversified
11	2011	0.1956	Moderately diversified
12	2012	0.1991	Moderately diversified

	year	hhi	classification
13	2013	0.1950	Moderately diversified
14	2014	0.1956	Moderately diversified
15	2015	0.2136	Moderately diversified
16	2016	0.2168	Moderately diversified
17	2017	0.2147	Moderately diversified
18	2018	0.2171	Moderately diversified
19	2019	0.2211	Moderately diversified
20	2020	0.2367	Moderately diversified
21	2021	0.2174	Moderately diversified
22	2022	0.2124	Moderately diversified
23	2023	0.2170	Moderately diversified
24	2024	0.2135	Moderately diversified

The Ecuador extension uses seven mutually exclusive branches: agriculture, mining and utilities, manufacturing, construction, trade and hotels, transport and communication, and other activities. This matters because a three-sector HHI has a theoretical floor of ($1/3 = 0.333$), already above the 0.25 threshold for the least-diversified category. With seven branches, the floor falls to ($1/7$), restoring discriminatory power.

	period	minimum_hhi	maximum_hhi	latest_hhi
0	2000-2024	0.1824	0.2367	0.2135

Ecuador remains moderately diversified over 2000-2024. The highest HHI is 0.2367 in 2020, consistent with a shock year, but the series does not cross the 0.25 least-diversified threshold.

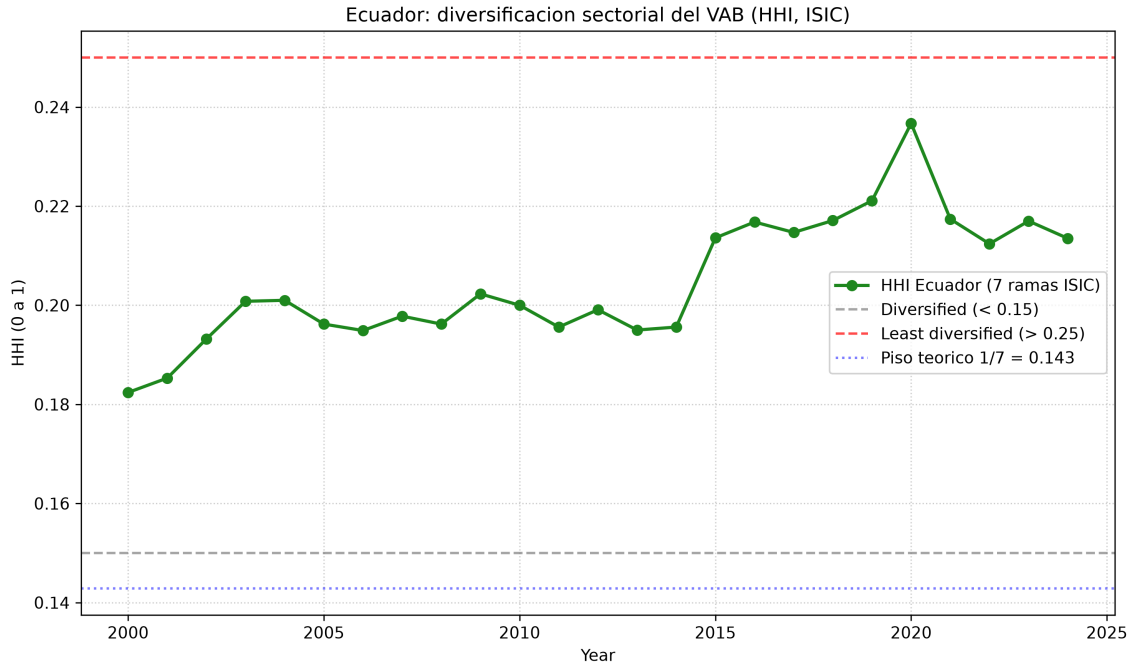


Figure 1: Ecuador HHI trend

7.2 DEA for Ecuador

	model	n_inputs	n_outputs	n_dmus	threshold_3x	verdict	n_efficient	mean_efficiency
4	Ecuador_Diversification_DEA	3	3	25	12	VALIDO	1	0.4857
5	Ecuador_Sector_Transition_DEA	7	6	7	6	VALIDO	1	0.6331

Two Ecuador DEA extensions are estimated. The first uses years as DMUs and is kept as an exploratory time-series design. Mining and utilities share is the input, while three branch shares are outputs: manufacturing, transport and communication, and trade, restaurants and hotels. The model asks whether each year reaches a productive and service-sector branch mix with lower reliance on the extractive/utilities branch.

	factor	efficiency_ccr_input	status
0	2000	0.756964	Optimization terminated successfully. (HiGHS S...
1	2001	1.000000	Optimization terminated successfully. (HiGHS S...
2	2002	0.789841	Optimization terminated successfully. (HiGHS S...
3	2003	0.654007	Optimization terminated successfully. (HiGHS S...
4	2004	0.444859	Optimization terminated successfully. (HiGHS S...
5	2005	0.330177	Optimization terminated successfully. (HiGHS S...
6	2006	0.252186	Optimization terminated successfully. (HiGHS S...

	factor	efficiency_ccr_input	status
7	2007	0.227706	Optimization terminated successfully. (HiGHS S...
8	2008	0.191196	Optimization terminated successfully. (HiGHS S...
9	2009	0.335150	Optimization terminated successfully. (HiGHS S...
10	2010	0.259741	Optimization terminated successfully. (HiGHS S...
11	2011	0.225982	Optimization terminated successfully. (HiGHS S...
12	2012	0.232633	Optimization terminated successfully. (HiGHS S...
13	2013	0.255841	Optimization terminated successfully. (HiGHS S...
14	2014	0.291619	Optimization terminated successfully. (HiGHS S...
15	2015	0.627644	Optimization terminated successfully. (HiGHS S...
16	2016	0.721653	Optimization terminated successfully. (HiGHS S...
17	2017	0.607293	Optimization terminated successfully. (HiGHS S...
18	2018	0.546072	Optimization terminated successfully. (HiGHS S...
19	2019	0.568149	Optimization terminated successfully. (HiGHS S...
20	2020	0.709220	Optimization terminated successfully. (HiGHS S...
21	2021	0.532207	Optimization terminated successfully. (HiGHS S...
22	2022	0.456062	Optimization terminated successfully. (HiGHS S...
23	2023	0.557324	Optimization terminated successfully. (HiGHS S...
24	2024	0.569626	Optimization terminated successfully. (HiGHS S...

The year-level model has 25 DMUs and an adequacy threshold of 12, so it passes the same screening rule applied to the UAE audit. It identifies one efficient year and a mean efficiency of 0.4857.

The strengthened Ecuador design uses sectors as DMUs. This is preferable to using years alone because the decision units are comparable branches observed from the same national accounts source. To keep the model valid with seven sectors, it uses one input and one output: average initial value-added share over 2000-2004 and average recent value-added share over 2020-2024.

	factor	initial_share	recent_share	efficiency_ccr_input	status	share_change	share_ratio
0	Construc- tion	0.02604	0.04322	1.000000	Optimiza- tion ter- minated success- fully.	0.01718	1.659754

		factor_initial_share	share_recent_efficiency	share_crr_input	status	share_change	share_ratio
					(HiGHS S...		
1	Other ac- tivities	0.29236	0.37862	0.780264	Optimiza- tion ter- minated success- fully. (HiGHS S...	0.08626	1.295047
2	Mining & utilities	0.08588	0.10558	0.740706	Optimiza- tion ter- minated success- fully. (HiGHS S...	0.01970	1.229390
3	Trade, res- taurants & hotels	0.16284	0.17386	0.643272	Optimiza- tion ter- minated success- fully. (HiGHS S...	0.01102	1.067674
4	Agri- culture, forestry & fishing	0.10648	0.08798	0.497820	Optimiza- tion ter- minated success- fully. (HiGHS S...	-0.01850	0.826258
5	Manufac- turing	0.20830	0.13830	0.400027	Optimiza- tion ter- minated success- fully. (HiGHS S...	-0.07000	0.663946

	factor	initial_share	recent_share	efficiency	CCR_input	status	share_change	share_ratio
6	Transport & communication	0.11812	0.07242	0.369395	Optimization terminated successfully. (HiGHS S...		-0.04570	0.613105

The sector-level model has seven DMUs and an adequacy threshold of six, so it also passes the $(3(m+s))$ rule. Construction lies on the frontier because its recent share is much larger than its initial share. Other activities and mining and utilities also gain relative weight. Manufacturing and transport and communication are below the frontier because their recent shares are lower than their initial shares. This is not a full knowledge-economy DEA, but it is a stronger sectoral transition exercise than the year-only model.

8 Discussion

The HHI replication is successful and classification-preserving. The UAE result supports the descriptive claim that fixed capital formation was moderately diversified in 2010-2017. The DEA replication is more critical: the universal-efficiency result appears when the model has too many dimensions for the available DMUs. Once the design is constrained by an adequacy rule, the frontier regains discriminatory power.

The Ecuador extension reinforces the importance of measurement design. HHI values depend on the sectoral partition. A three-sector partition has a floor above the least-diversified threshold and therefore cannot answer the policy question cleanly. A seven-branch partition produces a more informative result: Ecuador is moderately diversified, with a visible 2020 concentration increase but no regime change in classification. The sector-level DEA adds a second layer: moderate aggregate diversification can coexist with uneven sectoral transition, especially when manufacturing and transport lose relative weight.

9 Limitations

The UAE HHI comparison depends on source versioning and may reflect rounding or later revisions to sectoral fixed-capital-formation data. The DEA exercises are diagnostic, not structural causal models. The sector-level Ecuador DEA is stronger than the year-only version because the DMUs are economic branches, but it still uses value-added shares only. A fuller knowledge-economy DEA would need sectoral or provincial variables on employment, skills, innovation, infrastructure, exports, productivity, or firm dynamics.

10 Conclusion

The replication confirms the UAE HHI result but qualifies the DEA evidence. HHI is stable across the replicated and published values, while DEA efficiency is highly sensitive to the ratio between

DMUs and variables. The Ecuador extension shows that the same logic can be transferred if the sectoral aggregation is fine enough to preserve variation. The strengthened sector-level DEA suggests that Ecuador's diversification path is not uniform across branches.

11 Next Steps

The immediate next step is to enrich the sector-level Ecuador DEA with branch variables beyond value-added shares. The most useful additions would be employment, exports, productivity, capital formation, firm creation, digital adoption, or innovation by branch. A provincial DEA remains attractive if comparable administrative data can be assembled for all provinces, but the current sector-level model is the stronger reproducible extension available from the existing official source.